

Computational Complexity & Explanations in Physics - Scott Aaronson (UT Austin)

- Originally designed for philosophy.

Computational Complexity

Study of the inherent resources (time, memory, etc) needed to solve computational problems, and how those resources scale with problem size.

Others include - Randomness, Quantumness

are resources too.

Scaling behavior - hardness/ease $\propto$ chip/length

Computational intractability - (Sources)

- limited information
- chaotic sensitive dependence
- irrationality / bias.

F. d' 300 Bc CcD

① Quantum Computing interpretation

① To explain how we get such huge speedups, to posit the reality of

Blog

Probability theory with
Double slit experiment

P_1 = Probability of ph

P_2 = Same as with

both holes = probability

— 1930 - Computability Systematized.

Chaos - Infinitely fast computer - Chaos
— weather prediction will require modeling every atom.

→ Alan Turing - Imitation games.

→ "Why philosophers should care about Computational Complexity."

LHC cluster to analyze particle collisions.

→ half of world's economy AI build out
AI Doing science/physics.

3 Case Studies.

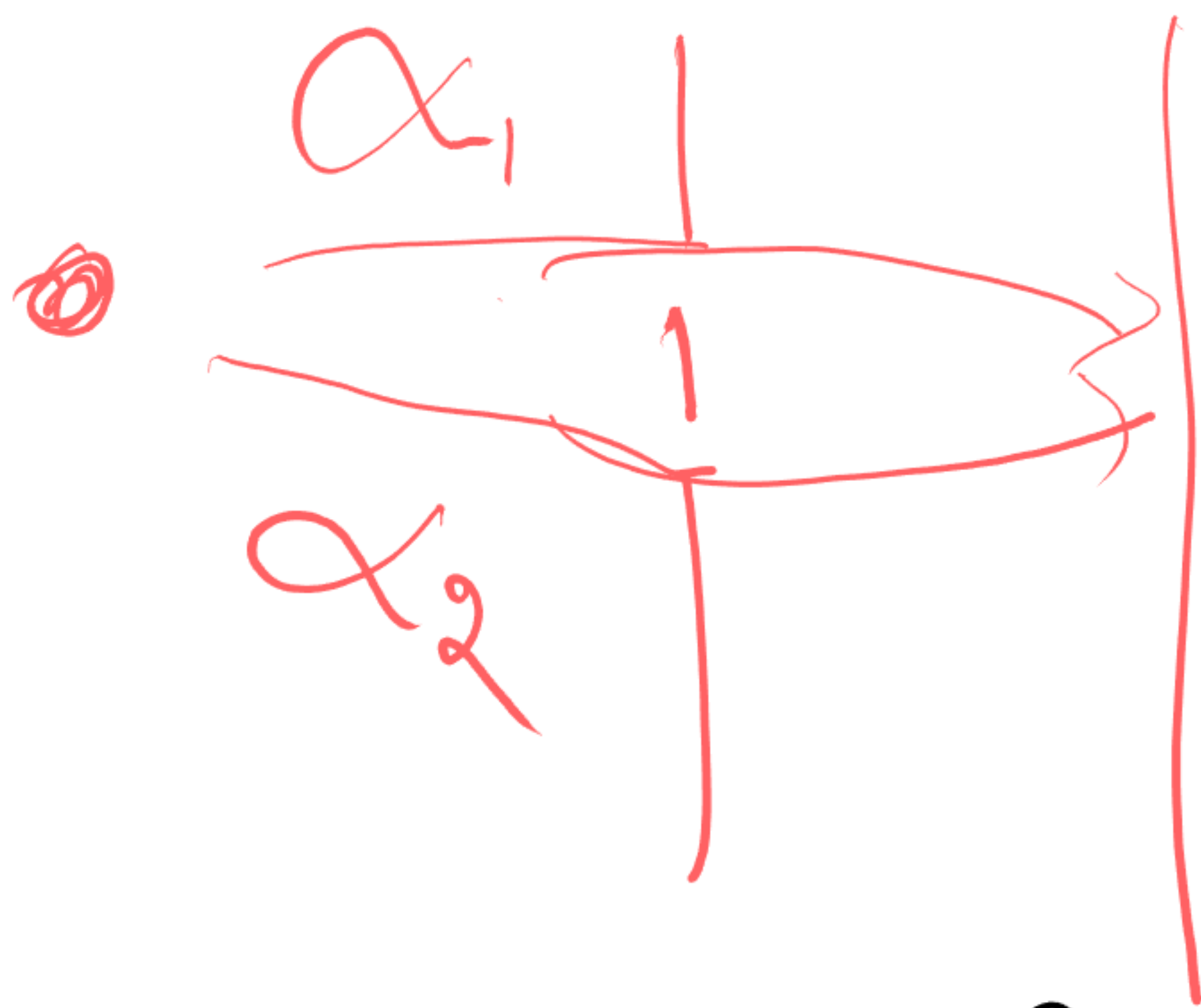
1) Quantum Computing & Many Worlds

2) Complexity Theory & Occam's Razor

3) Complexity Theory & AI for FA div

- at exploration

To explain



$$\alpha_1 = \cos \theta_1$$

$$\alpha_2 = \sin \theta_2$$

$$P = |\alpha|^2$$

$$\alpha_1 = \frac{1}{2} \quad \& \quad \alpha_2 = -\frac{1}{2}$$

By decreasing number
we increase probability

God prefers complex

Theories

② Daniel Harlow & Patrick

→ Region of validity of
quantum field theory

Relativity.

if $P = NP \Rightarrow$ Maxwell's da

2-norm over 1-norm

Hayden

What is quantum

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle$$

qubit - bit that are

basic building

x are orthogonal

dimension space

Taking a linear combination

& Many Worlds

fun Computers
we forced
parallel universes

minus signs
of

for a spot.

only second whole.

$P_1 + P_2$

it doesn't give

plex numbers
field amplitudes

$\alpha_1 + \alpha_2$

$$\alpha = \alpha_1 + \alpha_2 = 0$$

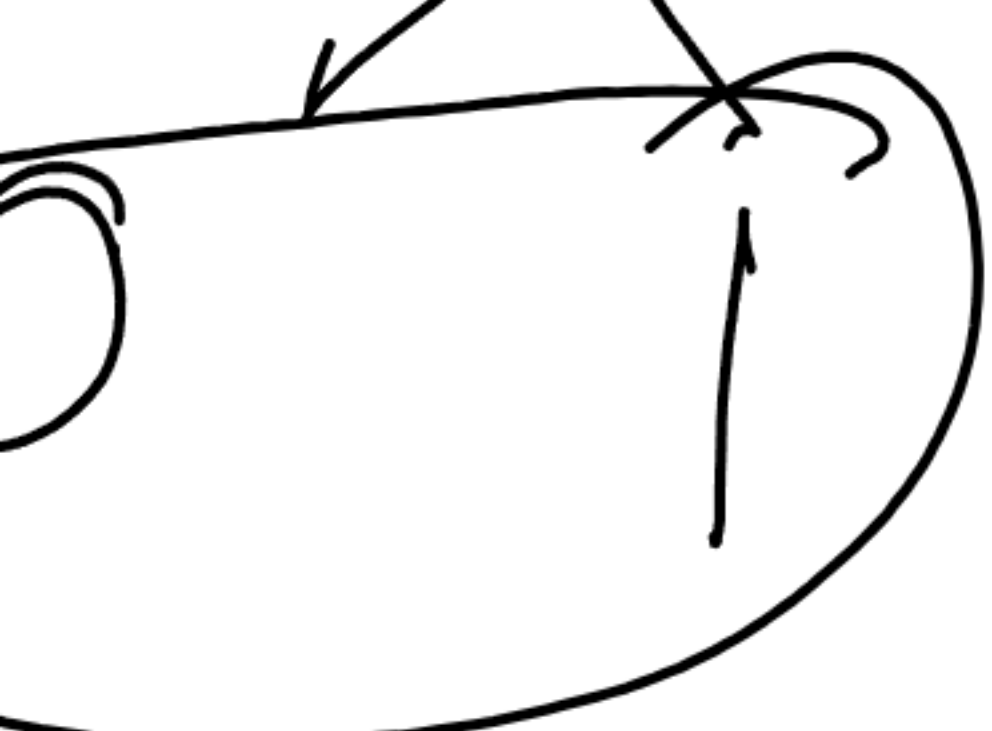
$$P|\alpha|^2 = 0$$

of paths

bers &

①

Computing

plitude of α
 Superpositi

vectors in 2^n
asis
e.

tion of basis

veers

Key point - P describe
- I need two am

- qbit is in Superpos
until I make a m

Born Rule

if I have 2 qbits I
 $\{00, 01, 11, 10\} \Rightarrow 2^n$

so 2^n amplitudes. for n -

If you need a vector
to keep track of ω

a Qubit

plitudes.

$\{0, 1\}$

ition of $\{0, 1\}$
~~or more~~

need 4 amplitudes

every qubit needs 2 or more

qubits

of 2^n complex numbers

that n-particles are

doing. Each particle encodes
We need 2^{1000} am

This is staggering amount
for nature

Classical Computers get
of quantum states to store

You never directly see the
When I see n -qubits - I see
nature to make a de
Outcome it gives me
Probability of each
1/2

1 qbit)
plitudes to description
+ of computation

overwhelmed by size
or manipulate

amplitudes
nbits $\uparrow$ force
cision about which

Out com

$$P = \frac{1}{N}$$

Main Known Applications

- ① Breaking current public
Shor's algorithm will →
- ② Simulating Quantum physics
- ③ Everything else — AI
Grover's algorithm

Ewin Tang's PhD thesis
Machine Learning With
Quantum — Breakthrough
Dequantization.

ions

Public-Key Cryptography,
the an upgrade

is

optimization

Quantum
at Any
gh in